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Authenticated Gateway-Assisted DNSSEC for IoT Clients.

Anshuman Katyan Jha [Student] - anshuman.katyanjha.cse28@heritageit.edu.in
Anusmriti Sinha Roy [Student] - anusmriti.sinharroy.iotcs28@heritageit.edu.in
Jhalak Dutta [Faculty] - jhalak.dutta@heritageit.edu

AIORI Hackathon 3

Problem Description

IoT devices need to trust DNS responses, but lack the resources to verify them independently.

- **Trust Gap:** Traditional DNS does not inherently authenticate responses, enabling spoofing and malicious redirection.
- **Resource Gap:** DNSSEC verification demands processing power and memory that constrained IoT devices may lack.
- **Security Gap:** DNS integrity alone is insufficient; the device must also authenticate its gateway and secure the communication channel.
- **Validation Gap:** Devices need a reliable way to distinguish authentic, unsigned, and invalid DNS data.
- **Reliability Gap:** DNS failures and repeated queries can increase latency and resource consumption.

Its necessary to move complex validation to a trusted gateway without compromising DNS security

- **Offload Verification:** Let the gateway perform DNSSEC validation using Unbound.
- **Secure the Connection:** Use HTTPS/TLS with gateway certificate verification.
- **Enforce Trust Policies:** Return validated answers, handle unsigned domains by policy, and reject bogus responses.
- **Reduce Device Overhead:** Minimize DNSSEC processing on the ESP32.
- **Measure Effectiveness:** Evaluate security, latency, and resource usage.

Lightweight IoT devices should not have to choose between limited resources and trustworthy DNS resolution.

Background

The DNS (Domain Name System) translates domain names into IP addresses, enabling not just IoT devices but other clients, like web browsers, email clients, servers, data centers as well to communicate with cloud services. However, conventional DNS does not inherently authenticate such responses, making the users vulnerable to DNS spoofing and response manipulation. Here is where DNSSEC (Domain Name System Security Extensions) comes into play, which addresses these issues by using digital signatures and a chain of trust to authenticate DNS data and detect invalid responses.

However, implementing DNSSEC validation directly on resource-constrained IoT devices can introduce much more computational and memory overhead than required. Here is where gateway-assisted DNSSEC offers an optimisation, by delegating the validation to a gateway while allowing the client to communicate with it through an authenticated, encrypted connection using TLP (Transport Layer Protocol). Thus, this approach aims to reduce client-side processing requirements while preserving DNS validation outcomes, as necessary.

Literature & RFC Review

[1] **RFC 4033 - DNS Security Introduction and Requirements** - March 2005
It introduces DNSSEC, explaining its purpose, capabilities and limitations and establishes that DNSSEC provides data-origin authentication and data integrity through digital signatures, although does not provide confidentiality. It also defines the important validation states : secure, insecure, bogus, and indeterminate, which must be correlated with our gateway's validation results correctly.
Relevance : Conceptual foundation for delegating DNSSEC validation to a trusted gateway, while securing the communication channel
[2] RFC 4035 - Protocol Modifications for DNSSEC - March 2005
It specifies how DNSSEC operates in practice, including signed-zone requirements, resolver behaviour, and the validation of DNS responses and explains how the DNSSEC resource records are used to authenticate DNS data and how validating resolvers handle invalid responses.
Relevance : Primary protocol reference for implementing and testing the gateway's DNSSEC validation.
[3] RFC 7858 - Specification for DNS over Transport Layer Security (TLS) - May 2016
It specifies DNS over TLS (DoT), which protects DNS queries and responses in transit. It describes TLS session establishment, server authentication considerations, connection reuse, and performance overhead. It also discusses the additional latency, CPU processing and me
Relevance : Background on encrypted DNS transport. Our client uses HTTPS over TLS to communicate with the gateway API, rather than using DoT directly.
[4] **RFC 9520 - Negative Caching of DNS Resolution Failures** - December 2023
It addresses the caching of DNS resolution failures including validation ones which would require resolvers to cache the failures under specified conditions to reduce unnecessary repeated queries and upstream DNS traffic
Relevance : Gateway's failure handling and performance evaluation.
[5] **Unbound DNSSEC Resolver Documentation**
Unbound is a validating DNS resolver which supports DNSSEC validation and configurable trust anchors. Its documentation provides practical guidance on resolver configuration, DNSSEC behaviour, and diagnostic procedures.
Relevance : Gateway implementation and communication, and reporting of validation outcomes, threat modelling, and resource benchmarking

Proposed Solution

A trusted gateway that delivers DNSSEC-verified resolution to resource-constrained IoT devices.

- IoT Client: ESP32-WROOM-32 requests DNS resolution without performing DNSSEC cryptographic verification locally.
- Authenticated Gateway: Fedora Linux runs FastAPI and secures client communication using HTTPS over TLS 1.2/1.3 with certificate-chain and hostname verification.
- DNSSEC Validation: Unbound validates DNS signatures and the chain of trust.
- Trust-Based Responses: Classify results as SECURE, INSECURE or BOGUS; reject bogus answers and apply explicit policies to unsigned domains in a controlled DNS Lab.
- Resilient Gateway: Access controls, rate limiting, structured logging, and negative caching improve reliability.
- Performance Evaluation: Compare against conventional unvalidated UDP DNS using repeated latency, CPU, and memory measurements.

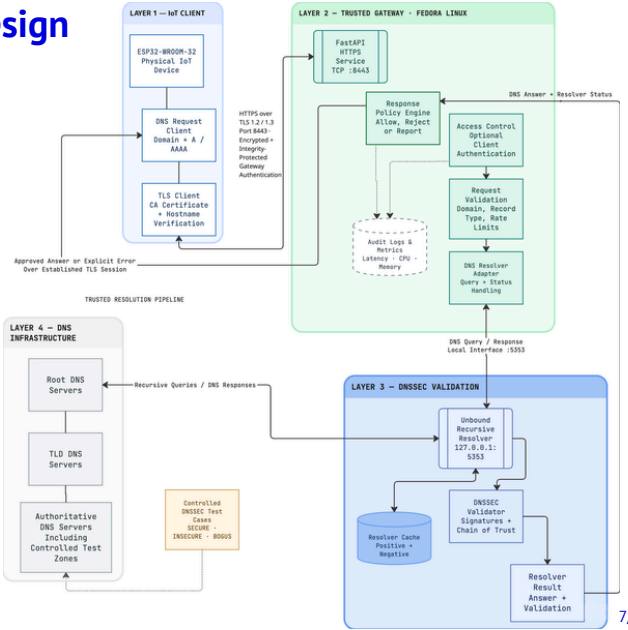
Quantifying the benefits and resource trade-offs by shifting DNSSEC verification to a trusted gateway while keeping IoT clients lightweight and communication authenticated.

Optimization Proposed

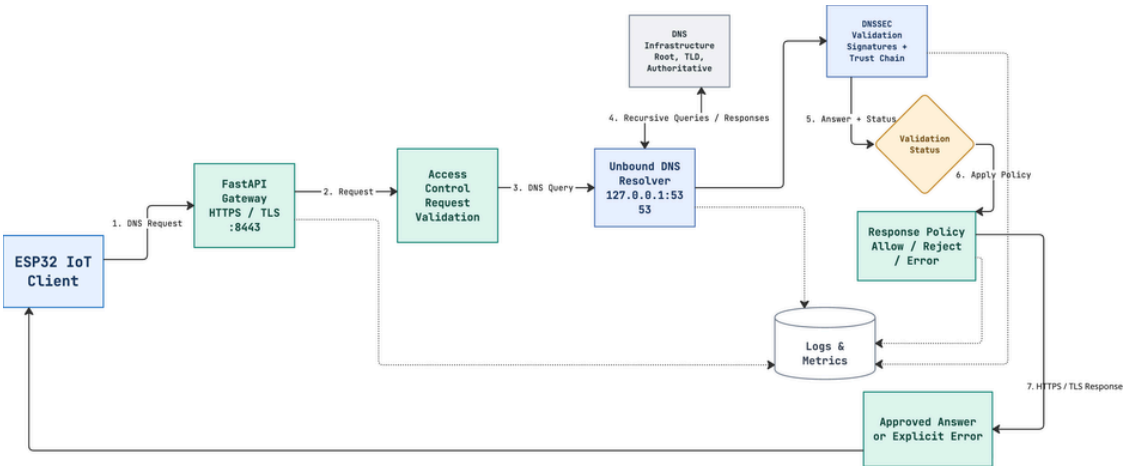
OPTIMISATION AREA	BEFORE	AFTER	EXPECTED BENEFITS
DNSSEC Validation	DNS responses accepted without DNSSEC validation	Unbound validates signatures and chain of trust at gateway	DNS authenticity; DNS spoofing mitigation
Secure Communication	Conventional DNS over UDP without transport encryption	UDP without transport encryption HTTPS over TLS 1.2/1.3 with CA and hostname verification	Confidentiality, integrity, gateway authentication
Resource Utilization	Client-side DNSSEC validation adds CPU and memory overhead	Validation delegated to gateway	Reduced IoT client workload
Response Handling	No explicit distinction between secure, insecure, and bogus results	Status classification; reject bogus responses; configurable insecure-domain policy	Reliable, policy-driven DNS decisions
Gateway Security	Potential gateway impersonation and unrestricted requests	Certificate verification, access control, input validation, rate limiting	Reduced attack surface; stronger access security
Failure Management	Repeated failed queries; limited failure handling	Timeouts, bounded retries, negative caching, structured logging	Improved resilience; reduced redundant queries
Performance Evaluation	No systematic benchmarking	Measure latency, gateway CPU/memory, and ESP32 free heap	Quantifiable security–performance trade-offs
Testing	Variable results from public DNS testing	Controlled SECURE, INSECURE, and BOGUS test zones	Repeatable tests; verifiable security outcomes

Solution Architecture & Design

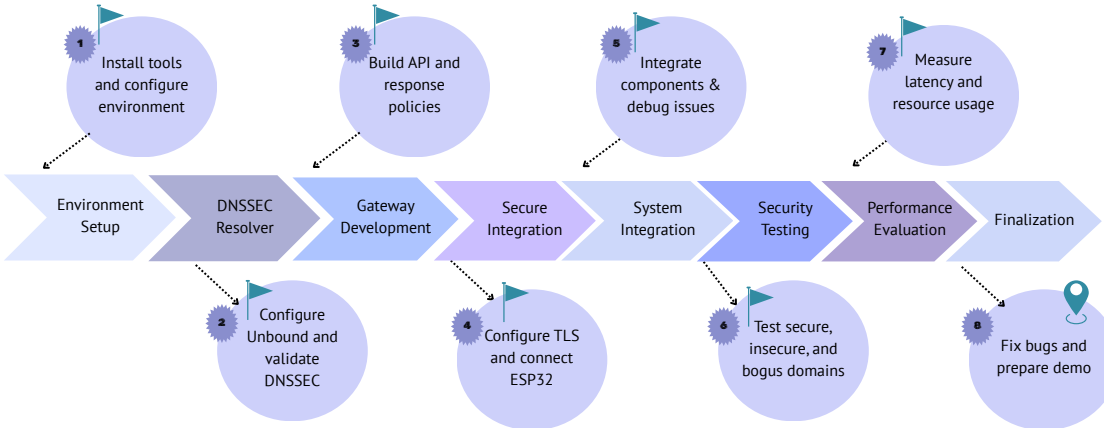
1. **The ESP32 sends DNS requests over HTTPS to the gateway.**
2. **Unbound validates DNSSEC, and the gateway applies policy: accept SECURE, handle INSECURE according to policy, and reject BOGUS.**
3. **Performance is evaluated using latency, CPU, and memory usage.**



Data Flow Diagram



Deployment Timeline



Bibliography I

[1] Arends, R., Austein, R., Larson, M., Massey, D., & Rose, S. (2005). *DNS Security Introduction and Requirements*. RFC 4033. Internet Engineering Task Force (IETF).

[2] Arends, R., Austein, R., Larson, M., Massey, D., & Rose, S. (2005). Protocol Modifications for the DNS Security Extensions. RFC 4035. Internet Engineering Task Force (IETF).

[3] Hu, Z., Zhu, L., Heidemann, J., Mankin, A., Wessels, D., & Hoffman, P. (2016). Specification for DNS over Transport Layer Security (TLS). RFC 7858. Internet Engineering Task Force (IETF).

[4] Fujiwara, K., & Kato, A. (2023). Negative Caching of DNS Resolution Failures. RFC 9520. Internet Engineering Task Force (IETF).

[5] NLnet Labs. (n.d.). Unbound Documentation.

[6] FastAPI. (n.d.). FastAPI Documentation.

[7] Espressif Systems. (n.d.). ESP-IDF Programming Guide.

[8] Internet Engineering Task Force. (2013). DNS Cookies. RFC 7873.

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Thank You for Your Attention !